

# Bio Informatics

## Lecture 6

# Needleman–Wunsch Algorithm – Example 2

- This algorithm can be used for any two strings.
- This guide will use two small DNA sequences as examples as shown below:
- Sequence (1): ACTGATTCA
- Sequence (2): ACGCATCA
- The length is different. The first step is to construct a grid.

# Needleman–Wunsch – Example 2

[illegible]

# Needleman–Wunsch Algorithm - Scores & Functions

- Match: +2
- Mismatch: -3
- Gap: -2

$$f(i, j) = \max \begin{cases} f(i-1, j-1) + S(x_i, y_i) \\ f(i-1, j) + S(\text{Gap}) \\ f(i, j-1) + S(\text{Gap}) \end{cases}$$

$$f(i, j) = \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |   | A | C | T | G | A | T | T | C | A |
|---|---|---|---|---|---|---|---|---|---|---|
|   | 0 | ? |   |   |   |   |   |   |   |   |
| A |   |   |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |   |   |
| G |   |   |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |   |   |
| A |   |   |   |   |   |   |   |   |   |   |
| T |   |   |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |   |   |
| A |   |   |   |   |   |   |   |   |   |   |

Match: +2; Mismatch: -3; Gap: -2;

$$f(i, j) = \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} \dots \\ \dots \\ 0 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |        | A | C | T | G | A | T | T | C | A |
|---|--------|---|---|---|---|---|---|---|---|---|
|   | 0 ← -2 | ? |   |   |   |   |   |   |   |   |
| A |        |   |   |   |   |   |   |   |   |   |
| C |        |   |   |   |   |   |   |   |   |   |
| G |        |   |   |   |   |   |   |   |   |   |
| C |        |   |   |   |   |   |   |   |   |   |
| A |        |   |   |   |   |   |   |   |   |   |
| T |        |   |   |   |   |   |   |   |   |   |
| C |        |   |   |   |   |   |   |   |   |   |
| A |        |   |   |   |   |   |   |   |   |   |

Match: +2; Mismatch: -3; Gap: -2;

$$f(i, j) = \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} \dots \\ \dots \\ -2 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |   | A  | C  | T | G | A | T | T | C | A |
|---|---|----|----|---|---|---|---|---|---|---|
|   | 0 | -2 | -4 |   |   |   |   |   |   |   |
| A |   |    |    |   |   |   |   |   |   |   |
| C |   |    |    |   |   |   |   |   |   |   |
| G |   |    |    |   |   |   |   |   |   |   |
| C |   |    |    |   |   |   |   |   |   |   |
| A |   |    |    |   |   |   |   |   |   |   |
| T |   |    |    |   |   |   |   |   |   |   |
| C |   |    |    |   |   |   |   |   |   |   |
| A |   |    |    |   |   |   |   |   |   |   |

Match: +2; Mismatch: -3; Gap: -2;

$$f(i, j) = \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} \dots \\ \dots \\ \dots \end{cases}$$

# Needleman–Wunsch – Example 2

|   |   | A    | C    | T    | G    | A     | T     | T     | C     | A     |
|---|---|------|------|------|------|-------|-------|-------|-------|-------|
|   | 0 | ← -2 | ← -4 | ← -6 | ← -8 | ← -10 | ← -12 | ← -14 | ← -16 | ← -18 |
| A | ? |      |      |      |      |       |       |       |       |       |
| C |   |      |      |      |      |       |       |       |       |       |
| G |   |      |      |      |      |       |       |       |       |       |
| C |   |      |      |      |      |       |       |       |       |       |
| A |   |      |      |      |      |       |       |       |       |       |
| T |   |      |      |      |      |       |       |       |       |       |
| C |   |      |      |      |      |       |       |       |       |       |
| A |   |      |      |      |      |       |       |       |       |       |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} 0 + \overset{\dots}{(-2)} \\ \dots \end{cases}$$



# Needleman–Wunsch – Example 2

|   |                                                     | A | C | T | G | A | T | T | C | A |
|---|-----------------------------------------------------|---|---|---|---|---|---|---|---|---|
|   | 0 ← -2 ← -4 ← -6 ← -8 ← -10 ← -12 ← -14 ← -16 ← -18 |   |   |   |   |   |   |   |   |   |
| A | -2                                                  |   |   |   |   |   |   |   |   |   |
| C | ?                                                   |   |   |   |   |   |   |   |   |   |
| G |                                                     |   |   |   |   |   |   |   |   |   |
| C |                                                     |   |   |   |   |   |   |   |   |   |
| A |                                                     |   |   |   |   |   |   |   |   |   |
| T |                                                     |   |   |   |   |   |   |   |   |   |
| C |                                                     |   |   |   |   |   |   |   |   |   |
| A |                                                     |   |   |   |   |   |   |   |   |   |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} -2 + \dots \\ \dots \end{cases}$$

# Needleman–Wunsch – Example 2

|   |    | A  | C  | T  | G  | A   | T   | T   | C   | A   |
|---|----|----|----|----|----|-----|-----|-----|-----|-----|
|   | 0  | -2 | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2 |    |    |    |    |     |     |     |     |     |
| C | -4 |    |    |    |    |     |     |     |     |     |
| G |    |    |    |    |    |     |     |     |     |     |
| C |    |    |    |    |    |     |     |     |     |     |
| A |    |    |    |    |    |     |     |     |     |     |
| T |    |    |    |    |    |     |     |     |     |     |
| C |    |    |    |    |    |     |     |     |     |     |
| A |    |    |    |    |    |     |     |     |     |     |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} \dots \\ \dots \\ \dots \end{cases}$$

# Needleman–Wunsch – Example 2

|   |     | A  | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|----|----|----|----|-----|-----|-----|-----|-----|
|   | 0   | -2 | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | ?  |    |    |    |     |     |     |     |     |
| C | -4  |    |    |    |    |     |     |     |     |     |
| G | -6  |    |    |    |    |     |     |     |     |     |
| C | -8  |    |    |    |    |     |     |     |     |     |
| A | -10 |    |    |    |    |     |     |     |     |     |
| T | -12 |    |    |    |    |     |     |     |     |     |
| C | -14 |    |    |    |    |     |     |     |     |     |
| A | -16 |    |    |    |    |     |     |     |     |     |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} 0 + (2) \\ -2 + (-2) \\ -2 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |     | A  | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|----|----|----|----|-----|-----|-----|-----|-----|
|   | 0   | -2 | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | 2  | ?  |    |    |     |     |     |     |     |
| C | -4  |    |    |    |    |     |     |     |     |     |
| G | -6  |    |    |    |    |     |     |     |     |     |
| C | -8  |    |    |    |    |     |     |     |     |     |
| A | -10 |    |    |    |    |     |     |     |     |     |
| T | -12 |    |    |    |    |     |     |     |     |     |
| C | -14 |    |    |    |    |     |     |     |     |     |
| A | -16 |    |    |    |    |     |     |     |     |     |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} -2 + (-3) \\ -4 + (-2) \\ 2 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |     | A  | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|----|----|----|----|-----|-----|-----|-----|-----|
|   | 0   | -2 | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | 2  | 0  | ?  |    |     |     |     |     |     |
| C | -4  |    |    |    |    |     |     |     |     |     |
| G | -6  |    |    |    |    |     |     |     |     |     |
| C | -8  |    |    |    |    |     |     |     |     |     |
| A | -10 |    |    |    |    |     |     |     |     |     |
| T | -12 |    |    |    |    |     |     |     |     |     |
| C | -14 |    |    |    |    |     |     |     |     |     |
| A | -16 |    |    |    |    |     |     |     |     |     |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} -4 + (-3) \\ -6 + (-2) \\ 0 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |     | A  | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|----|----|----|----|-----|-----|-----|-----|-----|
|   | 0   | -2 | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | 2  | 0  | -2 | ?  | ?   | ?   | ?   | ?   | ?   |
| C | -4  | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| G | -6  | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| C | -8  | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| A | -10 | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| T | -12 | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| C | -14 | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |
| A | -16 | ?  | ?  | ?  | ?  | ?   | ?   | ?   | ?   | ?   |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} -4 + (-3) \\ -6 + (-2) \\ 0 + (-2) \end{cases}$$

# Needleman–Wunsch – Example 2

|   |     | A   | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|-----|----|----|----|-----|-----|-----|-----|-----|
|   | 0   | -2  | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | 2   | 0  | -2 | -4 | -6  | -8  | -10 | -12 | -14 |
| C | -4  | 0   | 4  | 2  | 0  | -2  | -4  | -6  | -8  | -10 |
| G | -6  | -2  | 2  | 1  | 4  | 2   | 0   | -2  | -4  | -6  |
| C | -8  | -4  | 0  | -1 | 2  | 1   | -1  | -3  | 0   | -2  |
| A | -10 | -6  | -2 | -3 | 0  | 4   | 2   | 0   | -2  | 2   |
| T | -12 | -8  | -4 | 0  | -2 | 2   | 6   | 4   | 2   | 0   |
| C | -14 | -10 | -6 | -2 | -4 | 0   | 4   | 2   | 6   | 4   |
| A | -16 | -12 | -8 | -4 | -5 | -2  | 2   | 1   | 4   | 8   |

Match: +2; Mismatch: -3; Gap: -2;

$f(i, j)$

$$= \max \begin{cases} f(\nwarrow) + S(\text{match OR mismatch}) \\ f(\uparrow) + S(\text{Gap}) \\ f(\leftarrow) + S(\text{Gap}) \end{cases}$$

$$? = \max \begin{cases} \dots \\ \dots \\ \dots \end{cases}$$

# Needleman–Wunsch Algorithm - Scores & Functions

|   |     | A   | C  | T  | G  | A   | T   | T   | C   | A   |
|---|-----|-----|----|----|----|-----|-----|-----|-----|-----|
| 0 |     | -2  | -4 | -6 | -8 | -10 | -12 | -14 | -16 | -18 |
| A | -2  | 2   | 0  | -2 | -4 | -6  | -8  | -10 | -12 | -14 |
| C | -4  | 0   | 4  | 2  | 0  | -2  | -4  | -6  | -8  | -10 |
| G | -6  | -2  | 2  | 1  | 4  | 2   | 0   | -2  | -4  | -6  |
| C | -8  | -4  | 0  | -1 | 2  | 1   | -1  | -3  | 0   | -2  |
| A | -10 | -6  | -2 | -3 | 0  | 4   | 2   | 0   | -2  | 2   |
| T | -12 | -8  | -4 | 0  | -2 | 2   | 6   | 4   | 2   | 0   |
| C | -14 | -10 | -6 | -2 | -4 | 0   | 4   | 2   | 6   | 4   |
| A | -16 | -12 | -8 | -4 | -5 | -2  | 2   | 1   | 4   | 8   |